MIL-M-20484C (MU)
10 February 1969
Superseding
MIL-M-20484B (MU)
18 June 1965

MILITARY SPECIFICATION

MOTOR, ROCKET, M4
LOADING, ASSEMBLING, MARKING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, marking, and packing for one type of rocket motor designated as Motor, Rocket, M4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-45460 - Propellant, T2
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:55Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY

7548183 - Box, Packing, Ammunition, for Jato M4 Assembly, Details and Packing.
7548416 - Rocket Motor, M4, Assembly.

PUBLICATIONS

ARMY

ET-7548416 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Loaded assemblies. - The loaded assemblies shall comply with all requirements specified on Drawing (dwg.) 7548416 and with all requirements in applicable specifications.

3.3 Propellant grain

3.3.1 Composition and material. - The propellant grain material and composition shall comply with the applicable requirements of Specification MIL-P-45460 except that the percentages listed are nominal and may be varied to meet the static functioning requirements. The manufacturer shall use no other ingredients other than those specified in MIL-P-45460. A minimum of 20 percent of reworked propellant shall be uniformly dispersed in each batch and shall not vary by more than plus or minus 5 percent in any batch. Rework material shall consist of clear scraps from previously finished propellant such as clippings, ends, mill chips, and whole or sectioned grains.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:55Z
Check the source to verify that this is the current version before use.

3.3.2 120 degrees centigrade heat test. - The propellant charge shall comply with the requirement of paragraph 3.4 of Specification MIL-P-45460.

3.3.3 Moisture content. - The mositure content of the propellant charge shall not exceed 0.5 percent.

3.4 Static functioning. - The motor assemblies shall not fail to ignite and shall comply with the following requirements when tested as specified in 4.4.1.

3.4.1 Metal parts. - The metal parts of the motor assembly sahll not break or become detached.

3.4.2 Action time

3.4.2.1 The action time at 0 degrees F. shall be not less than 0.43 seconds nor greater than 0.72 seconds and at 120 degrees F. The action time shall be no less than 0.28 seconds nor more than 0.44 seconds.

3.4.3 Maximum thrust

3.4.3.1 The maximum thrust at 0 degrees F. and at 120 degrees F. shall not exceed 600 pounds.

3.4.4 Total Impulse

3.4.4.1 The total impulse at 0 degrees F. shall be no less than 100 pound force seconds nor more than 150 pound force seconds and at 120 degrees F. the total impulse shall be no less than 105 pound force seconds nor more than 160 pound force seconds.

3.4.5 Ignition delay time. - The motor assemblies shall function in not less than 12.33 seconds nor more than 16 seconds.

3.5 Workmanship. - All parts shall be free of chips, dirt, grease rust, scale, and other foreign material. The cleaning method shall not be injurious to any of thyparts, nor shall the parts be contimainated by by the cleaning agents. Paintings and coatings on all components and assemblies shall be restored when such paintings and coatings have been marred or obliterated.

3.6 First Article Inspection - This specification makea provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.



Source: https://assist.dla.mil -- Downloaded: 202610-19T07:55Z
Check the source to verify that this is the currentversion before use.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 20 rocket motor assemblies and 20 propellant grains which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it



Source: https://assist.dla.mil -- Downloaded: 202610-19T07:55Z
Check the source to verify that this is the current version before use.

conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed - Assemblies and grains will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection - If any assembly or grain fails to comply with any of the applicable requirements, the First Article Sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or grain to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new First Article Sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with tis specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts of one interfix lot number from one manufacturer.



Source: https://assist.dla.mil -- Downloaded: 20260-19T07:55Z
Check the source to verify that this is the current version before use.

b. Igniters of one interfix lot number from one manufacturer.
c. Propellant from one lot number from one manufacturer.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performe don inspection lots of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section except where 100 percent inspection is specified, and be documented as part of the contractors inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Charge, Propelling (see dwg. 7548417).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Charge cracked or broken..... | Visual | 01001 |
| Major: | AQL 0.40 percent | | |
| 101. | Outside diameter..... | Gage | 01002 |
| 102. | Length..... | Gage | 01003 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5)..... | Visual | 01004 |

4.3.2.2 Loading Assembly, prior to assembly of nozzles (see dwg. 7548416).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Propellant grain broken or cracked..... | Visual | 02001 |
| 2. | Propellant grain missing..... | Visual | 02002 |
| Major: | AQL 0.40 percent | | |
| 101. | Powder retainer clip not fully engaged..... | Visual | 02003 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5)..... | Visual | 02004 |



Source: https://assist.dla.mil -- Downloaded: 20260-19T07:55Z
Check the source to verify that this is the current version before use.

4.3.2.3 Loading assembly, prior to assembly of nozzle plugs or cover (see dwg. 7548416).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Hydrostatic test stamp missing from either nozzle or payload adapter block | Visual | 03001 |
| 2. | Foreign matter in nozzle throat | Visual | 03002 |
| Major: | AQL 0.40 percent | | |
| 101. | Igniter assembly missing | Visual | 03003 |
| 102. | Nozzle loose or improperly seated | Visual/ Manual | 03004 |
| Minor: | AQL 1.50 percent | | |
| 201. | Sealing compound missing on nozzle plugs | Visual | 03005 |
| 202. | Sealing ring missing or damaged | Visual | 03006 |
| 203. | Evidence of poor workmanship (see 3.5) | Visual | 03007 |

4.3.2.4 Loaded Assembly (see dwg. 7548416).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Nozzle plug missing | Visual | 04001 |
| Minor: | AQL 2.50 percent | | |
| 201. | Total length of longer side, maximum (max.) | Gage | 04002 |
| 202. | Total length of shorter side, max | Gage | 04003 |
| 203. | Adhesive missing at tube thread | Visual | 04004 |
| 024. | Scratched or abraded area not painted | Visual | 04005 |
| 205. | Tag missing | Visual | 04006 |
| 206. | Marking misleading or unidentifiable | Visual | 04007 |



Source: https://assist.dla.mil -- Downloaded: 20260-19T07:55Z
Check the source to verify that this is the current version before use.

4.3.2.5 Wood packing box, prior to sealing (see dwg. 7548183).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.65 percent | | |
| 101. | Numbe/of rocket motors incorrect | Visual | 05001 |
| 102. | Sleeve missing | Visual | 05002 |
| 103. | Packing component missing | Visual | 05003 |
| Minor: None defined. | | | |

4.3.2.6 Sealed wood packing box (see dwg. 7548183).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 1.00 percent | | |
| 101. | Hardware missing, borken or loose | Visual/ Manual | 06001 |
| 102. | Strapping missing, broken or loose | Visual/ Manual | 06002 |
| 103. | ICC marking misleading or unidentifiable | Visual | 06003 |
| 104. | Contents exposed | Visual | 06004 |
| 105. | Board broken or split | Visual | 06005 |
| Minor: | AQL 2.50 percent | | |
| 201. | Hardware improperly engaged | Visual/ Manual | 06006 |
| 202. | Strapping improperly assembled | Visual/ Manual | 06007 |
| 203. | Car seal missing, unsealed or improperly positioned | Visual/ Manual | 06008 |
| 204. | Contents loose | Manual | 06009 |
| 205. | Marking misleading or uniden-tifiable | Visual | 06010 |



Source: https://assist.dla.mil -- Downloaded: 20260-19T07:55Z
Check the source to verify that this is the current version before use.

4.3.3 Testing

4.3.3.1 First Article Samples (see 4.2) - The samples submitted as the first article sample shall be tested as specified below after having been examined for all other drawing requirements. Rejection and re-submission shall be in accordance with 4.2.3.

4.3.3.1.1 Moisture content and 120 degrees C. heat test (see 3.3.4 and 3.3.5) Code Nos. 01005 and 01006.

4.3.3.1.2 Static functioning (see 3.4) - Code No. 04008

4.3.3.2 Production testing.

4.3.3.2.1 Moisture content and 120 degrees C. heat test. - For the analysis of 4.3.3.2.1.1 and 4.3.3.2.1.2 excess propellant shall be selected from 2 or more grains from each lot. The combined weight shall be not less than 150 grams with equal amounts from each grain.

4.3.3.2.1.1 Moisture content (see 3.3.3) - Major defect - Code No. 01005. - This analysis shall be performed as specified in Method 102.1 of Standard MIL-STD-286. Failure of the sampel to meet the requirement shall reject the lot.

4.3.3.2.1.2 120 degree C. heat test (see 3.3.2) - Major defect - Code No. 01006 - This analysis shall be performed as specified in Method No. 404.1 of Standard MIL-STD-286. Failure of the sample in meeting the requirement shall reject the lot.

4.3.3.2.2 Static functioning - Major defect - Code No. 04008.

4.3.3.2.2.1 0 degrees and 120 degrees F. - The sampling procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-1 code letter F, with an AQL of 0.65 percent for each limit and each temperature separately. If any motor fails to comply with the applicable static functioning requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.1 using the applicable test equipment 4.3.4.

4.3.4 Inspection equipment - Equipment Tabulation Number ET-7548416 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.



Source: https://assist.dla.mil -- Downloaded: 202610-19T07:55Z
Check the source to verify that this is the current version before use.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test Methods and procedures

4.4.1 Static functioning

4.4.1.1 0 degrees F. - The motor assemblies shall be conditioned at 0 degrees F. plus or minus 3 degrees F. for a period of 18 hours. Within five minutes after removal from the conditioning chamber, the assemblies shall be mounted to a suitably barricated test stand, connected to the equipment necessary to determine compliance with the static functioning requirements and fired.

4.4.1.2 120 degrees F. - The motor assemblies shall be tested as specified in 4.4.1.1 except that the conditioning temperature shall be 120 degrees F. plus or minus 3 degrees F.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A. - The rocket motors shall be packed in a wood fox in accordance with dwg. 7548183.

5.2 Marking - Marking of the boses shall be in accordance with dwg. 7548183.

5.3 Shipping - When rocket motors from more than one lot are shipped in the same carrier, each lot shall be kept separate. The division between lots must be clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.



Source: https://assist.dla.mil -- Downloaded: 202610-19T07:55Z
Check the source to verify that this is the currentversion before use.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL). - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.



Source: https://assist.dla.mil -- Downloaded: 20260-19T07:55Z
Check the source to verify that this is the current version before use.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five-digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND14. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1340



Source: https://assist.dla.mil -- Downloaded: 202610-19T07:55Z
Check the source to verify that this is the current version before use.